Short Proofs on Convex Sets

Convex Optimization, Class and Online Exercises

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Class exercise 1. Every affine set can be written as the solution set of a linear system. With $L = x_0 + V$ and $V = \{v : Pv = 0\}$, every $x \in L$ satisfies $Px = Px_0$, and the converse also holds.

Class exercise 2. I related the two descriptions of an ellipsoid, $\{x : (x - x_c)^T P^{-1}(x - x_c) \leq 1\}$ and $\{x_c + Au : \|u\|_2 \leq 1\}$. Substituting the second form into the first gives $A^T P^{-1} A = I$, which holds for $A = P^{1/2}$.

Class exercise 3. The perspective function $P(x, t) = x/t$ maps convex sets to convex sets: the image of a segment is again a segment, with a reparametrised θ. The inverse image of a convex set under the perspective is also convex.

Online exercise 1. A proof by contradiction. If C were not convex, some convex combination x_3 of two of its points would lie outside C, which contradicts the given property.

Online exercise 2. The Euclidean ball $\{x : \|x - x_c\|_2 \leq r\}$ is convex. The proof is one application of the triangle inequality to $\|\theta x_1 + (1 - \theta)x_2 - x_c\|_2$.